KYLMA-POS

Enterprise POS & Inventory Management System

System Documentation: User Guide, Installation, and Testing

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Source code and release	

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1. Introduction

KYLMA-POS is a desktop point-of-sale (POS) and inventory system for a small store. It runs on a Windows computer and stores all data in a Microsoft SQL Server database. It is a Windows program, not a website or a phone app.

The system lets the store do four things:

- Sell products to customers at the counter, with barcode scanning, discounts, tax, and cash change.
- Track products and stock, warn when stock is low, and record every stock change with a reason.
- Print and reprint receipts.
- Report daily sales, top products, and sales by category.

Prices and totals are shown in Philippine pesos (₱). Tax is 12% by default and is applied after any discount.

2. System Requirements

Item	Requirement
Operating system	Windows 10 or Windows 11
Runtime	.NET Framework 4.8
Database	SQL Server 2016 SP1 or later (SQL Server Express is sufficient)
Database tool	SQL Server Management Studio (SSMS)
Development (only to build from source)	Visual Studio 2022 with the ".NET desktop development" workload and the .NET Framework 4.8 Developer Pack
Printer	Any printer installed in Windows; an 80 mm thermal printer is recommended for receipts

3. Installation and Setup

3.1 Create the database

Run the database script once on the SQL Server. It creates the database named Mangubat, all tables, all stored procedures, and sample data. The script can be run again without causing harm.

1. Open SQL Server Management Studio (SSMS).
2. In the connection window, set Server name to .\SQLEXPRESS (or your server name), choose Windows Authentication, and click Connect.
3. Click File > Open > File, select 01_CreateDatabase.sql, and click Open.
4. Press F5 (or click Execute). Wait for the message "Commands completed successfully."

Verify the database. In a new query window, run:

```
SELECT name FROM sys.databases WHERE name = N'Mangubat';
USE Mangubat;  SELECT Username, RoleId, IsActive FROM dbo.Users;  SELECT COUNT(*) FROM dbo.Products;
```

You should see the Mangubat database, three users, and 14 products.

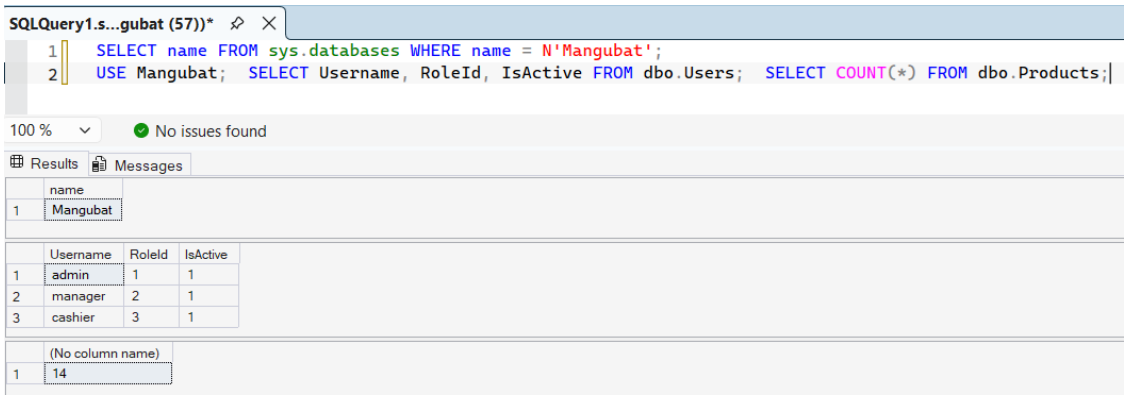


Figure 1. *Verification query results (database, users, products)*

3.2 Build the program

1. Open KYLMA-POS.slnx in Visual Studio.
2. In the toolbar, change Debug to Release.
3. Click Build > Rebuild Solution. Confirm there are zero errors.
4. The program files are in KYLMA.WinForms\bin\Release\.

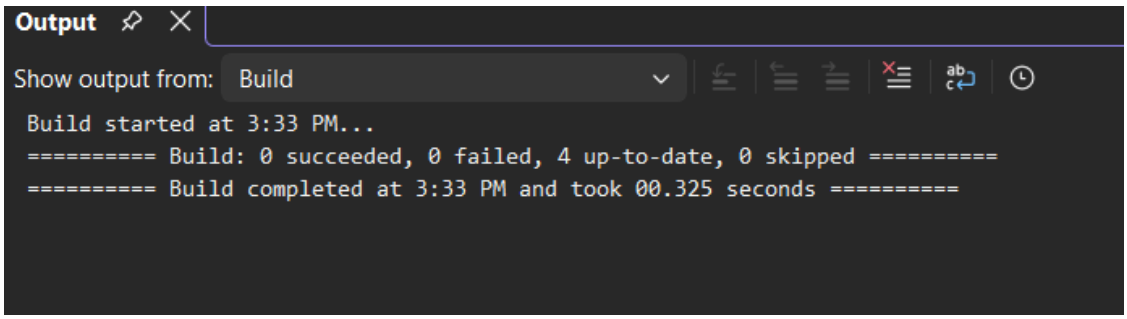


Figure 2. *Visual Studio Release build with zero errors*

3.3 Set the startup project

The startup project is the one that runs when you press F5. It must be KYLMA.WinForms.

1. In Solution Explorer, right-click KYLMA.WinForms.
2. Click Set as Startup Project.
3. Confirm that KYLMA.WinForms is shown in bold.

3.4 Connect the program to the database

Open KYLMA.WinForms\App.config (or KYLMA.WinForms.exe.config in the Release folder). Check that the Mangubat connection string points to your SQL Server. The default uses .\SQLEXPRESS with Windows Authentication. Also set the store name, address, phone, return policy, and tax rate in the appSettings section.

> **Note:** Do not share an App.config that contains real production passwords.

3.5 Start the program

Press F5 in Visual Studio, or double-click KYLMA.WinForms.exe in the Release folder. The login screen opens.

4. Roles and Accounts

The database has three starting accounts. Change these passwords before using real data.

Username	Password	Role	Access
admin	Admin@123	Admin	Everything, including Users & Roles
manager	Manager@123	Manager	Dashboard, POS, Inventory, Customers, Reprint Receipt, Sales Analytics
cashier	Cashier@123	Cashier	Dashboard, POS, Customers, Reprint Receipt

Each person should log in with their own account. Sales are recorded under the logged-in user.

5. Using the System

5.1 Logging in and the main screen

1. Type your username and password, then click Log in (or press Enter).
2. The Dashboard opens.
3. Use the menu on the left to open each screen. The menu shows only the items your role allows.
4. Press F2 at any time to open the Point of Sale.
5. Click Log out at the bottom left to close the program.

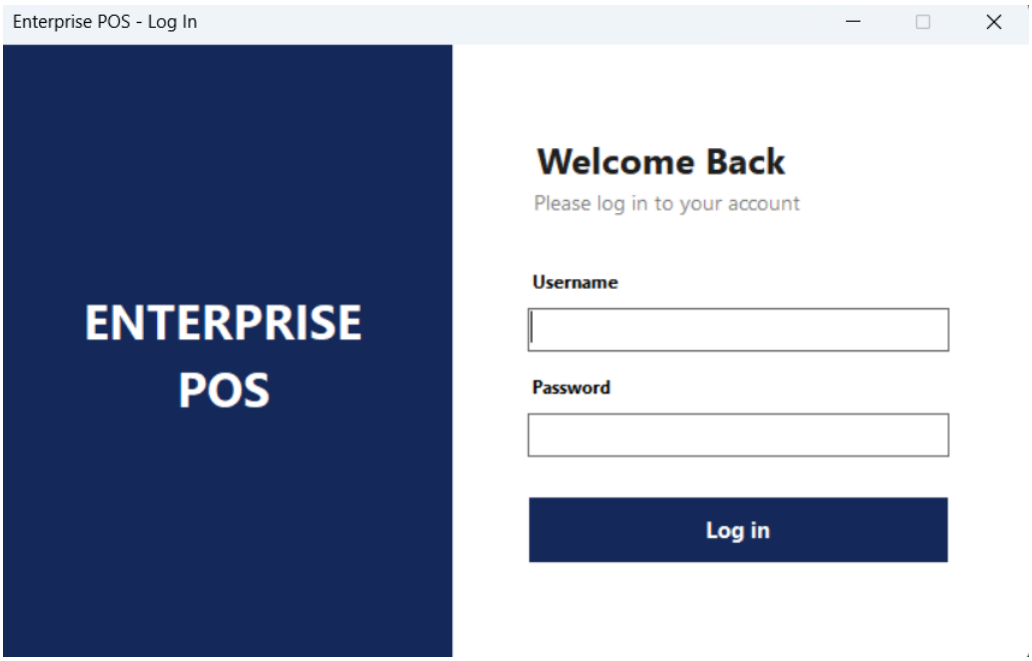


Figure 3. *Login screen*

If the username or password is wrong, the program shows a red message. It does not say which of the two was wrong, for security.

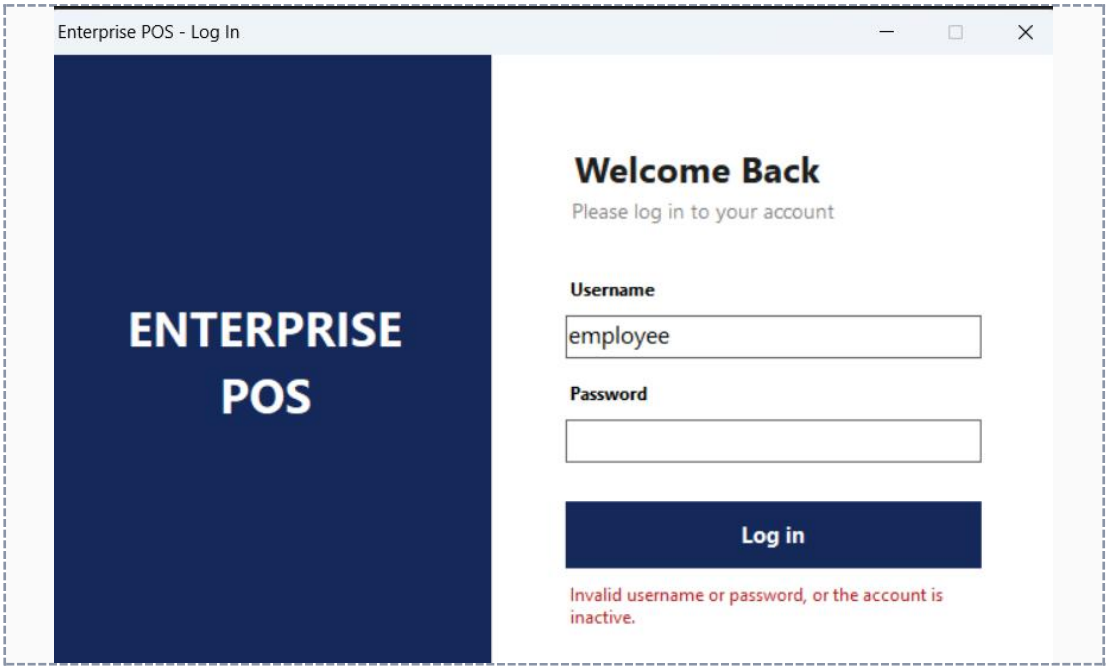


Figure 4. *Login error message for a wrong password or username*

The Dashboard shows four cards and a list:

- **Today's sales:** total money from completed sales today.
- **Orders today:** number of sales today.
- **Low-stock items:** products at or below their reorder level.
- **Active products:** products that can be sold.

The list below the cards shows the low-stock products.

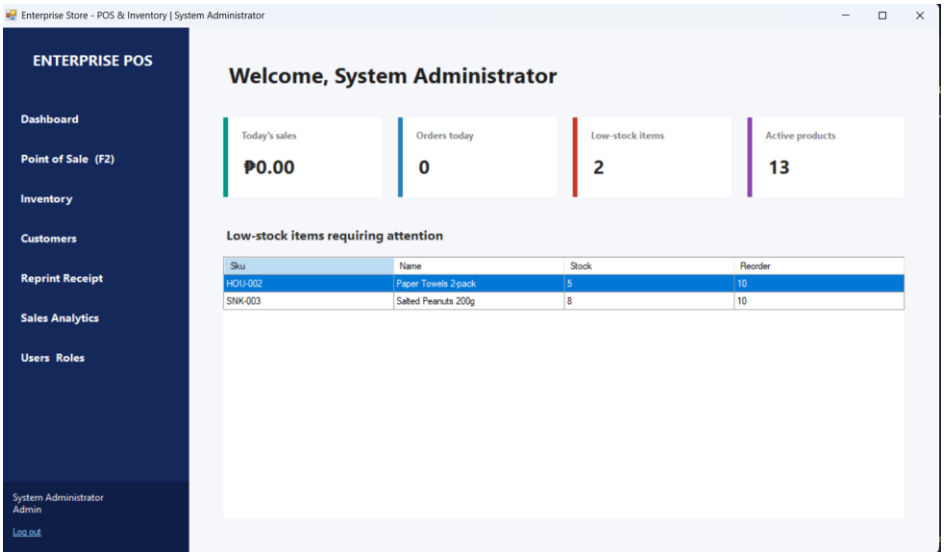


Figure 5. *Dashboard (Admin account) with metrics and low-stock list*

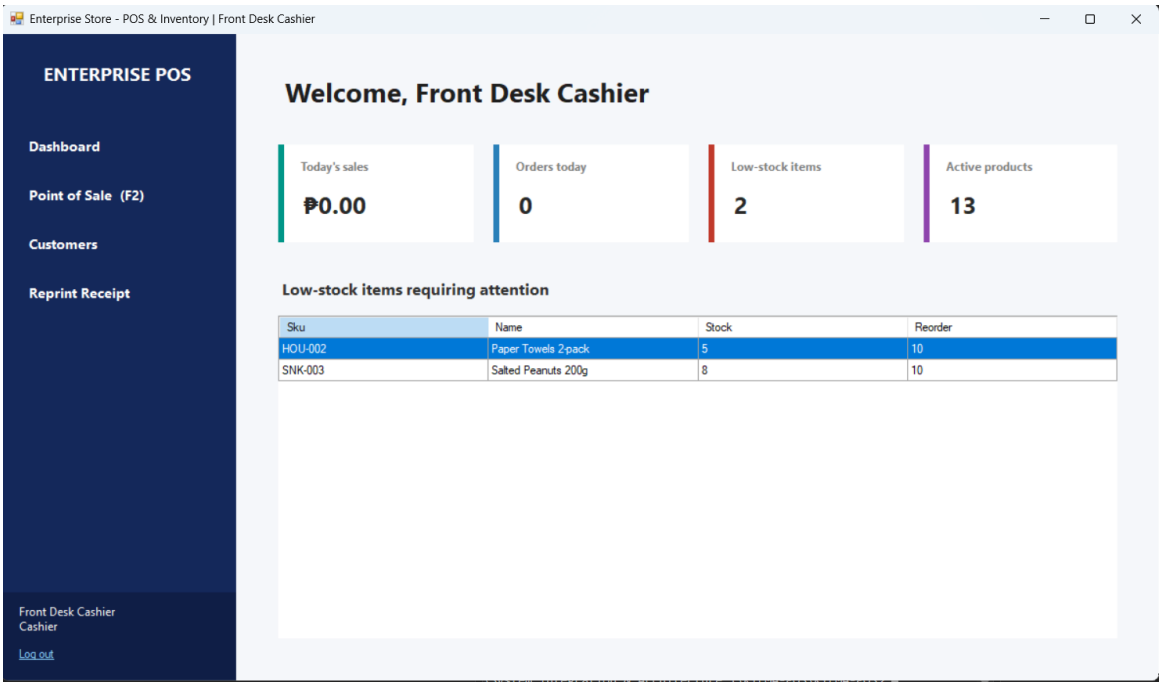


Figure 6. *Left menu for the Cashier account*

5.2 Inventory: adding and checking products

Open Inventory from the menu (Manager or Admin only).

- **Search:** type part of a SKU, barcode, or product name, then press Enter.
- **Show inactive:** tick to also see products removed from sale.
- **Red rows:** products at or below the reorder level. They need restocking.

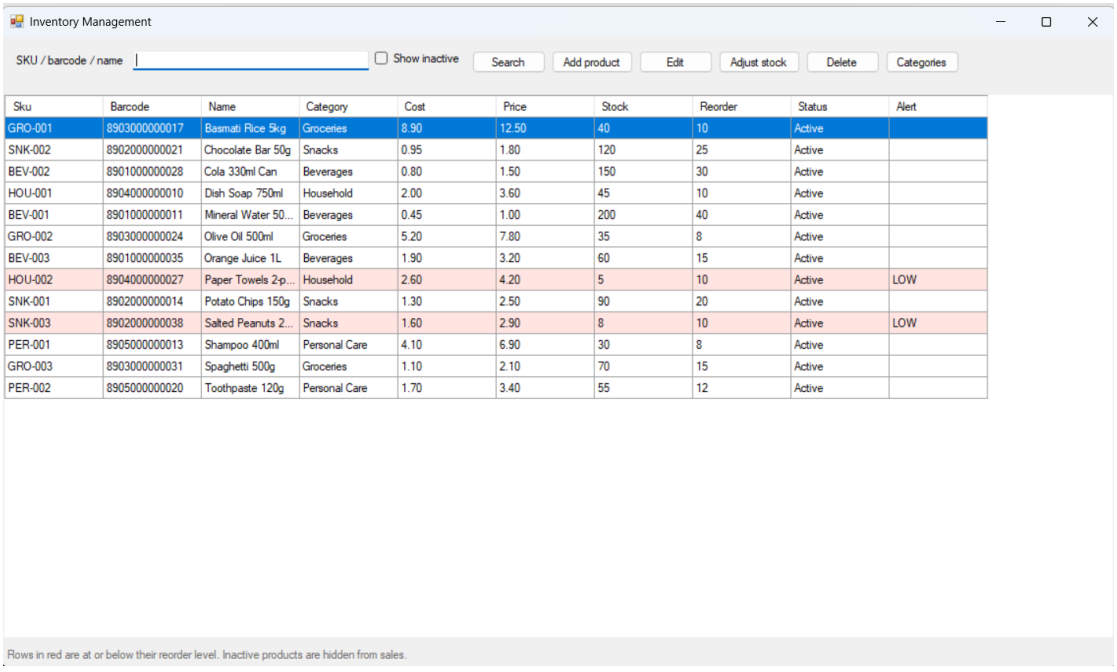


Figure 7. *Inventory screen with red low-stock rows*

Add a product

1. Click Add product.
2. Fill in the SKU (unique; letters, digits, - or _), the barcode (optional, unique), and the product name.
3. Choose a category.
4. Enter the cost price, selling price, opening stock, and reorder level.
5. Optionally choose an image (JPG or PNG, up to 2 MB).
6. Click Save.

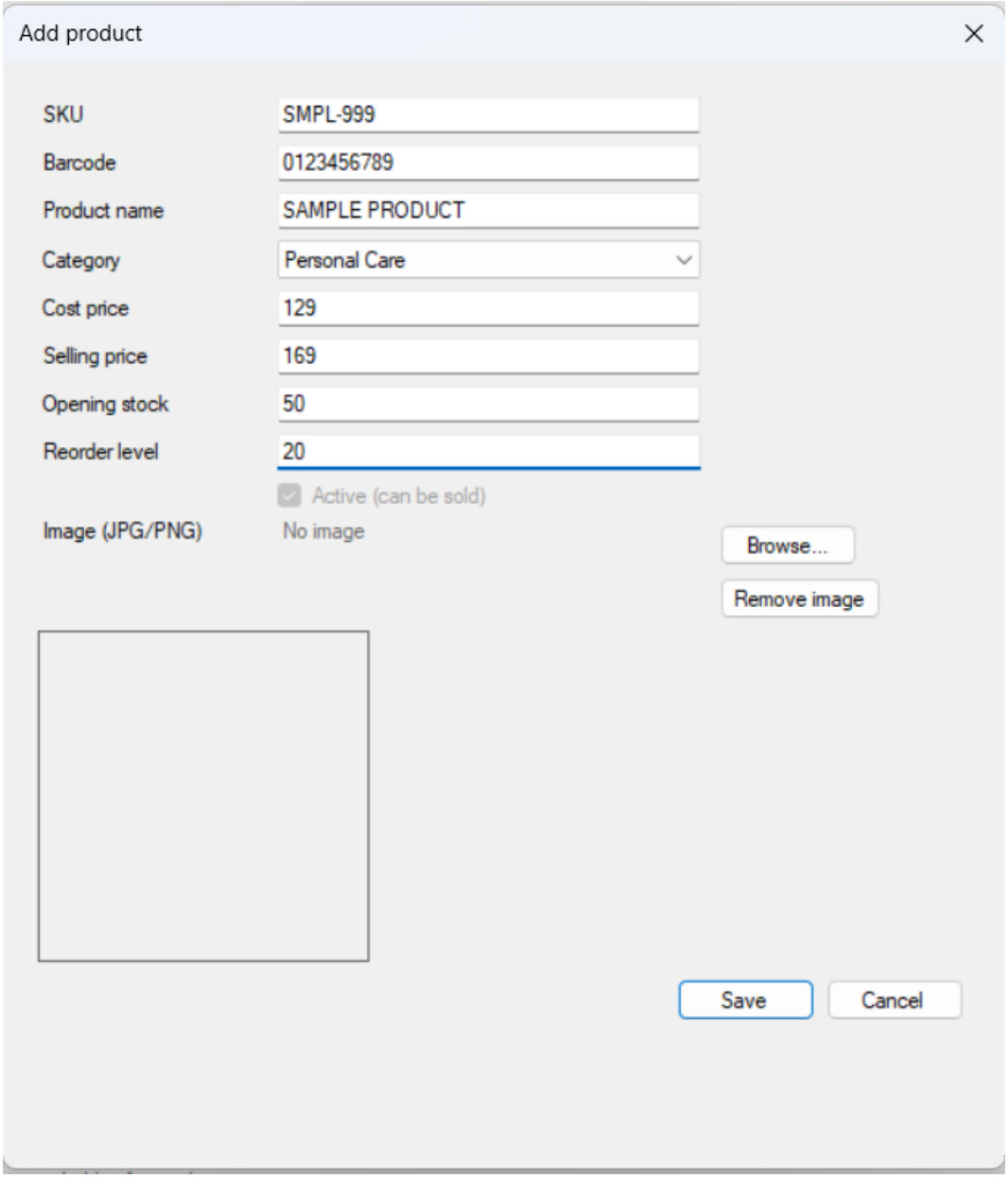
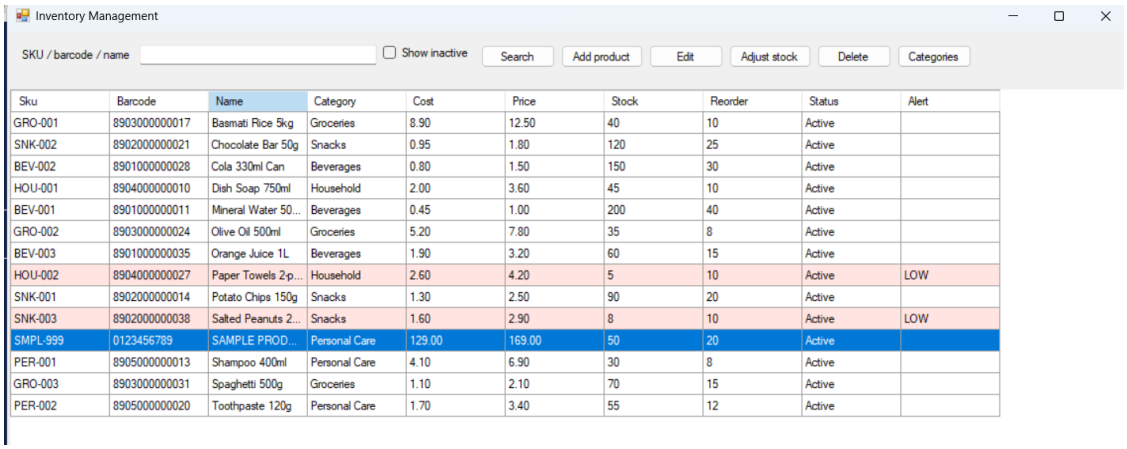


Figure 8. *Add product form filled in*



Sku	Barcode	Name	Category	Cost	Price	Stock	Reorder	Status	Alert
GRO-001	8903000000017	Basmati Rice 5kg	Groceries	8.90	12.50	40	10	Active	
SNK-002	8902000000021	Chocolate Bar 50g	Snacks	0.95	1.80	120	25	Active	
BEV-002	8901000000028	Cola 330ml Can	Beverages	0.80	1.50	150	30	Active	
HOU-001	8904000000010	Dish Soap 750ml	Household	2.00	3.60	45	10	Active	
BEV-001	8901000000011	Mineral Water 50...	Beverages	0.45	1.00	200	40	Active	
GRO-002	8903000000024	Olive Oil 500ml	Groceries	5.20	7.80	35	8	Active	
BEV-003	8901000000035	Orange Juice 1L	Beverages	1.90	3.20	60	15	Active	
HOU-002	8904000000027	Paper Towels 2-p...	Household	2.60	4.20	5	10	Active	LOW
SNK-001	8902000000014	Potato Chips 150g	Snacks	1.30	2.50	90	20	Active	
SNK-003	8902000000038	Salted Peanuts 2...	Snacks	1.60	2.90	8	10	Active	LOW
SMPL-999	0123456789	SAMPLE PROD...	Personal Care	129.00	169.00	50	20	Active	
PER-001	8905000000013	Shampoo 400ml	Personal Care	4.10	6.90	30	8	Active	
GRO-003	8903000000031	Spaghetti 500g	Groceries	1.10	2.10	70	15	Active	
PER-002	8905000000020	Toothpaste 120g	Personal Care	1.70	3.40	55	12	Active	

Figure 9. *Inventory list after saving the new product*

Edit a product

Select the product, click Edit, change the details, and click Save. The stock figure cannot be changed here. Use Adjust stock instead.

Delete a product

Select the product and click Delete, then confirm. The product is hidden from sales, but its history is kept. Tick Show inactive to see it again.

5.3 Fixing low stock

This is the most common daily task. Stock can only be changed with Adjust stock, which records every change.

1. On the Inventory screen, select the product (red rows need attention).
2. Click Adjust stock.
3. In Change (+ add, - remove), type a positive number to add stock or a negative number to remove it.
4. In Reason, write a short note (required), for example Supplier delivery, invoice 1024.
5. Click Apply adjustment. A message shows the new on-hand quantity.



Figure 10. *Adjust stock window before Apply*

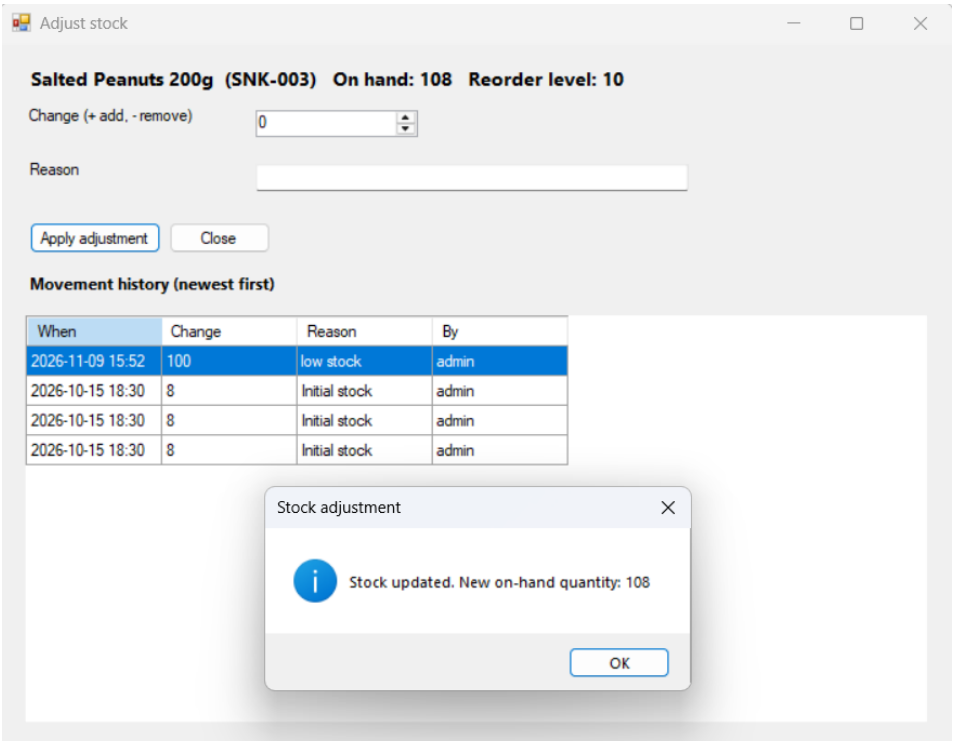


Figure 11. *Adjust stock result and movement history*

The Movement history at the bottom lists every change, with its date, amount, reason, and user.

> **Rules:** The system will not let stock go below zero. Once stock is above the reorder level, the row is no longer red.

Example: Cola 330ml has 25 cans and a reorder level of 30. A supplier delivers 100 cans. Select Cola, click Adjust stock, type +100, enter the reason "Supplier delivery", and click Apply adjustment. The new on-hand quantity is 125.

5.4 Point of Sale: making a sale

Open the Point of Sale with F2 or from the menu.

Step A: Add items

1. Click the SKU / barcode box.
2. Scan the barcode (scanners press Enter by themselves), or type the SKU or barcode and press Enter.
3. The item is added to the cart. The box clears for the next item.
4. To sell several of the same item at once, set the Qty box before scanning. For example, set Qty to 3, then scan.

Step B: Correct the cart

- **Change a quantity:** click the line, set Qty, then click Set qty of selected line.
- **Remove a line:** click the line, then click Remove selected line.
- **Start over:** click Clear cart and confirm.

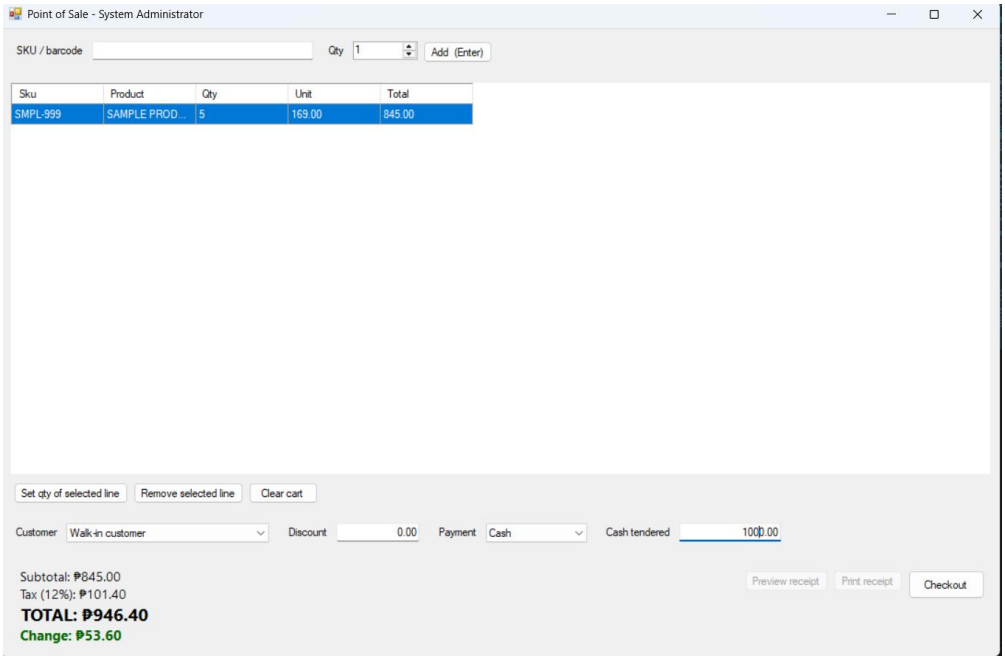


Figure 12. *Point of Sale with items in the cart*

Step C: Customer, discount, and payment

- **Customer:** leave as Walk-in customer, or choose a saved customer.
- **Discount:** type an amount of money (for example 5.00), not a percentage. It cannot be more than the subtotal.
- **Payment:** choose Cash, Card, or Mobile.
- **Cash tendered:** for Cash only. Type the amount received. The change appears in green. If the amount is too low, the message "Cash tendered is less than the total" appears and the sale is not completed.
- **Card and Mobile:** the exact total is used automatically.

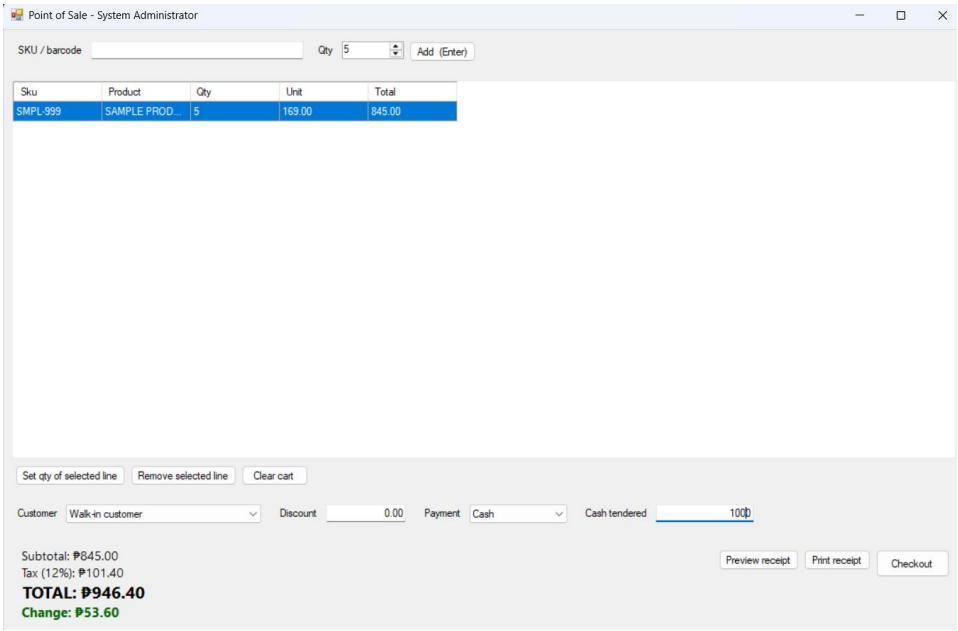


Figure 13. *Point of Sale showing cash change*

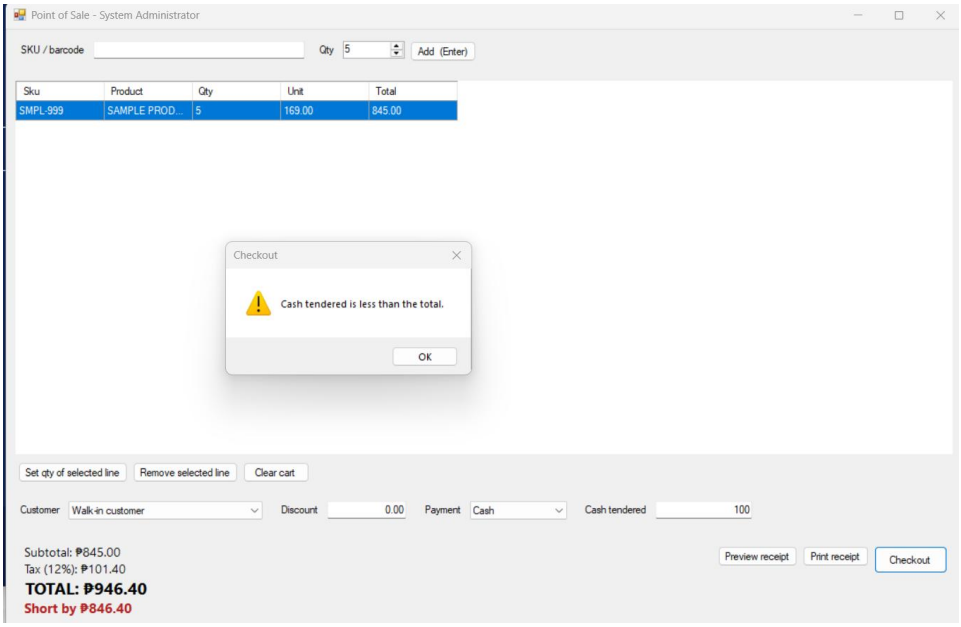


Figure 14. *Cash tendered too low: warning message*

Step D: Checkout

1. Click Checkout.
2. A message shows the order number (for example ORD-20261107-000012), the total, the amount paid, and the change.
3. Click Preview receipt or Print receipt to give the customer a receipt.
4. The cart clears automatically. Start the next sale.

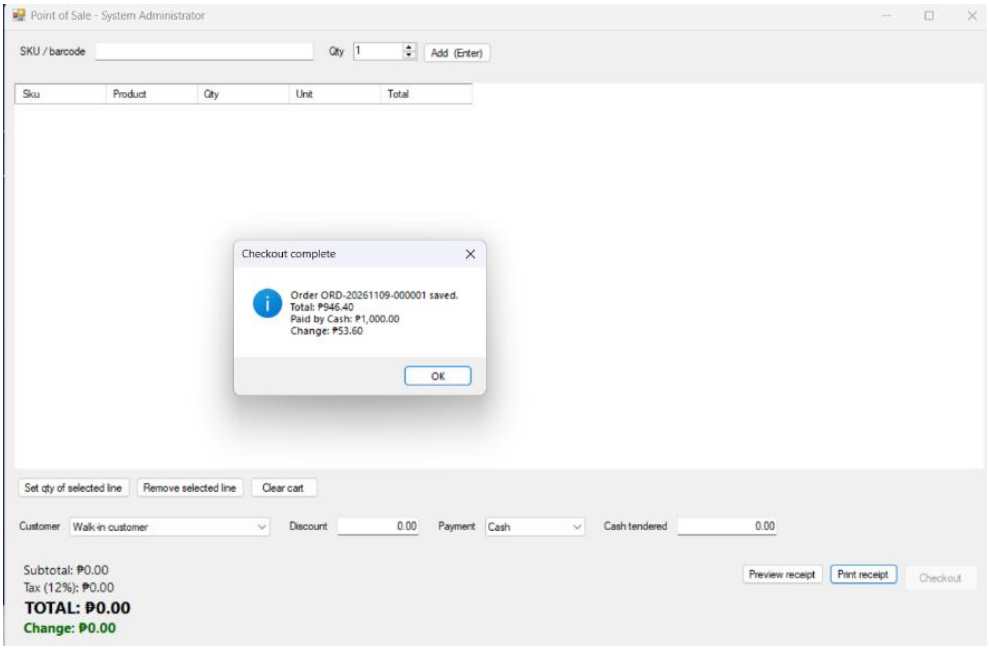


Figure 15. *Checkout complete message with order number*

> **Note:** Prices and stock are checked again in the database during checkout. Stock goes down exactly once for each completed sale.

5.5 Receipts

- **After a sale:** click Preview receipt or Print receipt.
- **Reprint later:** open Reprint Receipt from the menu, type the order number, and click Find. Then click Preview or Print. Reprinting never creates a new sale.

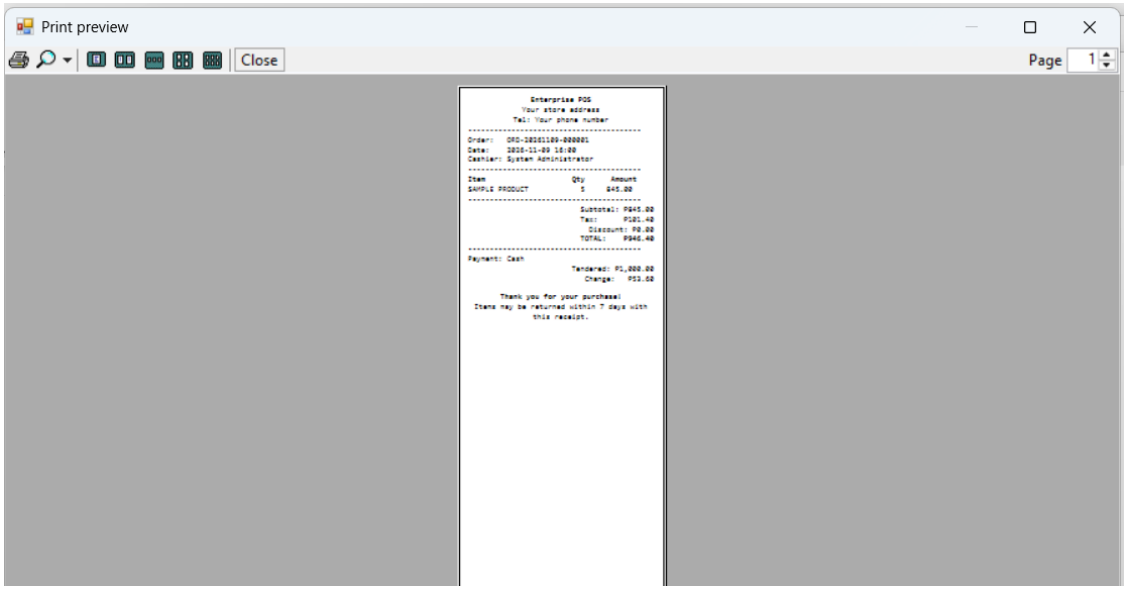


Figure 16. *Receipt print preview*

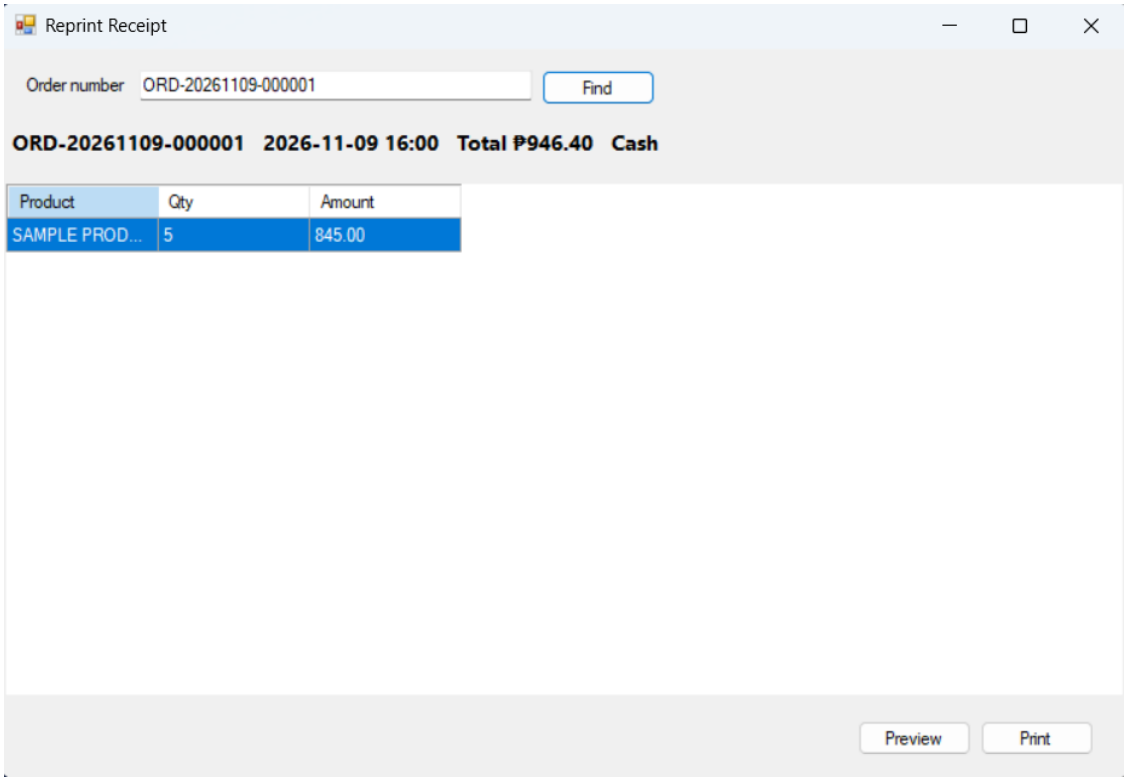


Figure 17. *Reprint Receipt screen with a found order*

5.6 Customers

Customers are optional. Use this screen to save regular buyers.

- Type a name or phone number and click Search to find a customer.
- To add a customer, enter the name (required) and optionally the phone, email, and address, then click Save customer.
- Saved customers appear in the Customer list on the Point of Sale screen.

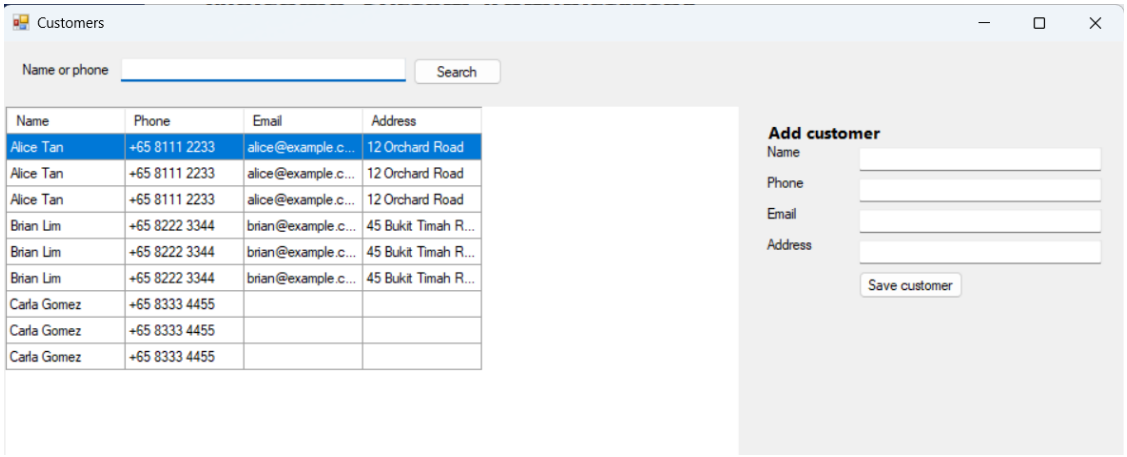


Figure 18. *Customers screen*

5.7 Sales Analytics (Manager and Admin)

1. Open Sales Analytics from the menu.
2. Choose the From and To dates.
3. Click Refresh.
4. Read the tabs: Sales by day, Top products, and By category.
5. Click Export daily sales (CSV) to save the daily table, which can be opened in Excel.

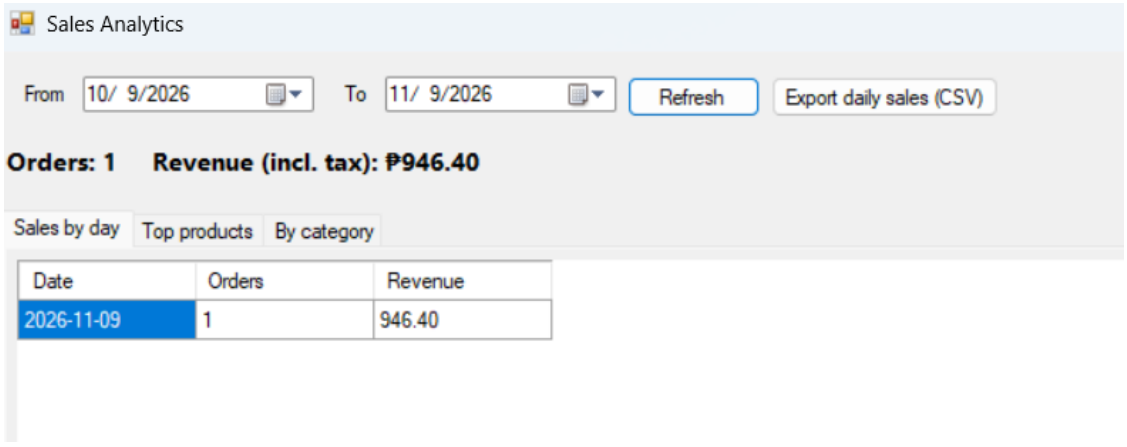


Figure 19. *Sales Analytics with daily results*

5.8 Users & Roles (Admin only)

- **Create a user:** enter a username, full name, password (at least 8 characters), and role, then click Create user.
- **Change a role or status:** select the user, change Role or Account active, then click Save role and status.
- **Reset a password:** select the user, click Reset password..., and type the new password.
- Admins cannot deactivate their own account or change their own role. Deactivated users cannot log in.

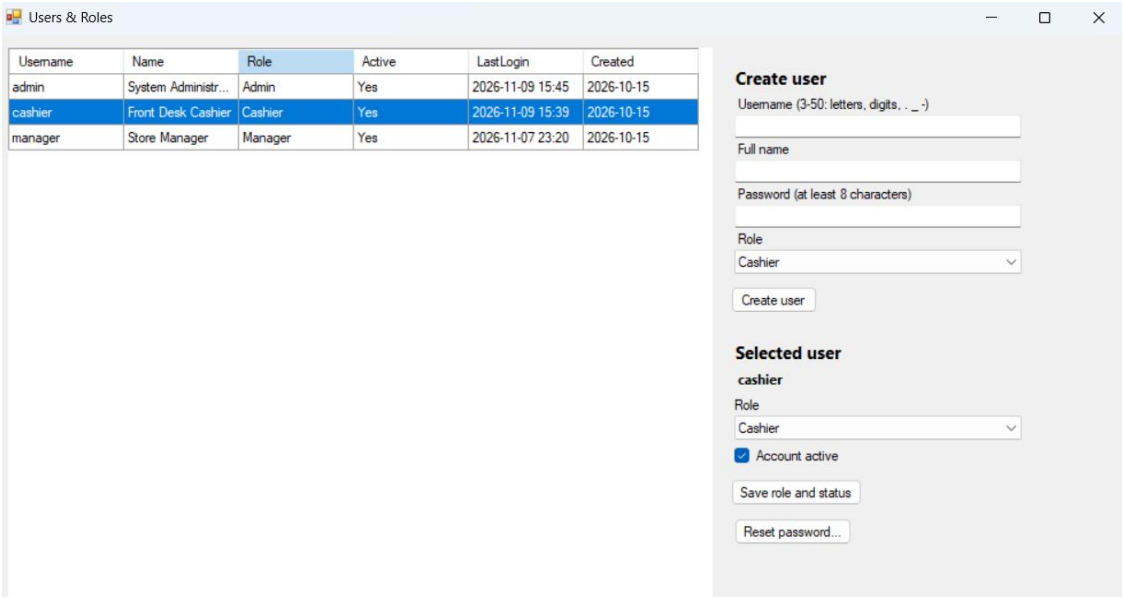


Figure 20. *Users & Roles screen*

5.9 A typical day

When	What to do
Opening	Log in with your own account. Check the low-stock list on the Dashboard.
During the day	Make sales in the Point of Sale (F2). Print receipts for customers.
When stock arrives	Inventory > select product > Adjust stock > enter the + quantity > reason > Apply adjustment.
When items are damaged or lost	Inventory > select product > Adjust stock > enter the - quantity > reason > Apply adjustment.
End of day or week	Sales Analytics > set dates > Refresh > Export CSV if a record is needed.

5.10 Common questions

Can I undo a sale? Not in this version. There is no void button. If a sale was wrong, correct the stock with Adjust stock, write a clear reason, and record the order number.

What happens if a price changes? Edit the product and change the selling price. New sales use the new price. Old receipts keep their original prices.

Why can't I see Inventory? Your role does not include it. Ask a manager or admin.

I forgot my password. Ask an admin to reset it under Users & Roles.

Why is a product not sold? It may be inactive. Tick Show inactive on the Inventory screen, then check the Active setting.

6. Database Design

The database is named Mangubat. Its main tables are:

Table	Purpose
Roles	The three roles: Admin, Manager, Cashier.
Users	Login accounts with salted password hashes, role, and active status.
Categories	Product categories.
Products	Products with SKU, barcode, prices, stock, reorder level, and optional image.
Customers	Optional saved buyers.
Orders	One row per sale: order number, totals, payment method, and amount paid.
OrderDetails	One row per item in each sale, with quantity and price at the time of sale.
InventoryLog	Every stock change (sales, deliveries, adjustments) with the reason and user.

Stored procedures handle sales, stock changes, and reports. Sales run inside a single transaction, so a failed sale changes nothing.

7. Testing

Run each test below after installation. Record the result and attach the screenshot listed where one is shown.

No.	Test	Expected result	Evidence	Result
1	Log in with correct details	Dashboard opens	Figure 5	☐ Pass Fail ☐
2	Log in with a wrong password or username	Error message; no access	Figure 4	☐ Pass Fail ☐
3	Log in as cashier	Only cashier menu items shown	Figure 6	☐ Pass ☐ Fail
4	Add a new product	Product appears in the list	Figure 9	☐ Pass ☐ Fail
5	Add stock with +100 and a reason	On-hand increases; row no longer red	Figure 11	☐ Pass ☐ Fail
6	Remove more stock than available	Refused with a message	Message text	☐ Pass ☐ Fail
7	Cash sale with change	Change shown; stock reduced once	Figure 13, Figure 15	☐ Pass ☐ Fail
8	Sell more than stock on hand	"Not enough stock" message	Message text	☐ Pass ☐ Fail
9	Cash below the total	Warning shown; sale not completed	Figure 14	☐ Pass ☐ Fail
10	Preview and print a receipt	Receipt matches the sale	Figure 16	☐ Pass ☐ Fail
11	Reprint by order number	Same receipt; no new sale	Figure 17	☐ Pass ☐ Fail
12	Add and find a customer	Customer saved and found	Figure 18	☐ Pass ☐ Fail
13	Sales Analytics totals	Totals match the test sales	Figure 19	☐ Pass ☐ Fail
14	Admin creates a user	New user can log in	Figure 20	☐ Pass ☐ Fail

8. Known Limitations

- There is no void or refund feature. Correct mistakes with Adjust stock and keep the order number.
- There is no audit table for user administration. Stock changes are recorded in InventoryLog.
- Passwords use salted SHA-256, which is acceptable for a school project but weaker than modern methods such as PBKDF2 or Argon2.
- Scanner support covers USB keyboard-type scanners. Serial and camera scanners need additional work.

- The program runs on one Windows computer or on several computers connected to the same SQL Server.

9. Troubleshooting

Problem	What to check
"Cannot reach the database server"	SQL Server service is running; the server name in App.config is correct; the firewall allows SQL Server.
Login always fails	Username and password are correct; the account is active; the database script was run.
Product list is empty	The database has active products; the logged-in account can read the database.
Receipt does not print	The printer is installed, online, and selected in Windows.
Amounts show dollars ($)	Use the version that sets the peso culture in Program.cs (en-PH) and rebuild.
Program will not start	.NET Framework 4.8 is installed; all DLL files are next to the .exe.

Error details are written to C:\ProgramData\KYLMA\Logs. Send this file to the administrator when reporting a problem.

10. Source Code and Release

The source code and database script are stored in the GitHub repository:

[GitHub repository link]

The ready-to-run program (KYLMA-POS-Release.zip) is attached to the release named v1.0 in that repository. To run it, extract the zip, run the database script, set the connection string in KYLMA.WinForms.exe.config, and open KYLMA.WinForms.exe.

Appendix A. Screenshot Guide

A.1 How to add the remaining screenshot

1. Open the login screen and type a wrong password (or a wrong username) with a correct one.
2. Click Log in so the red error message appears.
3. Press Windows + Shift + S, select the window, and save the image as PNG (for example login_error.png).
4. In Word, delete the dashed box in Figure 4 and use Insert > Pictures > This Device to add the image.
5. Set Wrap Text to In Line with Text and keep the caption under the picture.

A.2 Figure list

Figure	Screen	Section	Status
1	Verification query results (database, users, products)	3.1	Included
2	Visual Studio Release build with zero errors	3.2	Included
3	Login screen	5.1	Included
4	Login error message for a wrong password or username	5.1	To add
5	Dashboard (Admin account) with metrics and low-stock list	5.1	Included
6	Left menu for the Cashier account	5.1	Included
7	Inventory screen with red low-stock rows	5.2	Included
8	Add product form filled in	5.2	Included
9	Inventory list after saving the new product	5.2	Included
10	Adjust stock window before Apply	5.3	Included
11	Adjust stock result and movement history	5.3	Included
12	Point of Sale with items in the cart	5.4	Included
13	Point of Sale showing cash change	5.4	Included
14	Cash tendered too low: warning message	5.4	Included
15	Checkout complete message with order number	5.4	Included
16	Receipt print preview	5.5	Included
17	Reprint Receipt screen with a found order	5.5	Included
18	Customers screen	5.6	Included
19	Sales Analytics with daily results	5.7	Included
20	Users & Roles screen	5.8	Included